

XRPL Validator — Daily Engineering Report • 2026-08-06

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 7 commits (xrpl-validator 7)

Engineering changes

- **fix(probe): a dropped fetch is not a divergence — withhold the verdict** (cae6b85 , xrpl-validator) Both probes conflated "the network dropped this fetch" with "this object does not exist", so an incomplete pre-state replayed into a plausible, well-formed divergence. #106099077 is what prompted this: the scout filed two `tecPATH_DRY` vs `tesSUCCESS` circular-arb Payments on the FFI leg, with a triage note blaming the offer-book directory walk and `succ` /overlay merge. Re-running the same fixture on the same binary — ...
- **fix(ledger): a hydrated maker with no line for what it sells holds ZERO** (44d7a1c , xrpl-validator) #106096771 354/354 (was 352/354) and #106101763 224/224 (was 223/224) — three divergences across two independent ledgers, one root cause. Both came from the 2026-08-06 fresh scout batch, and #106101763 was never looked at while making the change: the first evidence this run that a fix generalises rather than fitting the ledger it came from.
- **fix(xrpl-ledger): auth is not an interest in a trust line** (f3f8b98 , xrpl-validator) rippled's `bLowReserveSet/bHighReserveSet` (`SetTrust.cpp:611-620`) are exactly quality-in, quality-out, `NoRipple-vs-DefaultRipple`, freeze, limit and balance. We also tested `lsfLowAuth/lsfHighAuth`, so a single authorisation bit pinned a zero-balance, zero-limit line open forever.
- **feat(xrpl-ledger): DX_SEL — measure strand selection against rippled's model** (dbb9110 , xrpl-validator) Settles what rippled actually does, and relocates the defect.
- **fix(xrpl-ledger): the bridge strand's upper bound must not charge the transfer fee** (005ad9a , xrpl-validator) rippled states the rule outright in `adjustQualityWithFees` (`BookStep.cpp`): "It is important that `qualityUpperBound` is an upper bound on the quality (it is used to ignore strands whose quality cannot meet a minimum threshold). When calculating quality assume no fee is charged, or the estimate will no longer be an upper bound." It therefore returns `ofrQ UNCHANGED` for a CLOB tip and for a multi-path AMM, and even in ...
- **fix(xrpl-ledger): strand selection — upper-bound order with fall-through** (03c2cb9 , xrpl-validator) The judge and the fall-through are ONE mechanism, which is why building the judge alone failed twice (recorded 8 v 17: it rejects the leading candidate and then crosses nothing).
- **fix(xrpl-ledger): a rejected candidate's offer removals must survive its rollback** (a836f94 , xrpl-validator) Hunted for a ledger exercising the `ofrsToRm` deviation noted in 03c2cb9. There isn't one: `DX_RM` swept all 144 staged fixtures (batch2, newfinds, freshfinds, corpus) and reported ZERO cases where a rejected candidate's rollback discarded a reap. The one hit, #105912454, is the benign kind -- offers the fill `CONSUMED`, which rippled discards too, since it drops the failed strand's sandbox and carries out only `ofrsToRm`.

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Auto-Expiring Trust Line Authorization** (compliance) Add an optional sfAuthExpiration (UInt32 ledger close-time) to RippleState, settable by the trust-line holder that granted authorization via a new tfS
- **Domain-Scoped Deposit Authorization** (compliance) Add sfDepositAuthDomain (Hash256, a PermissionedDomain keylet) to AccountRoot, settable via AccountSet when asfDepositAuth is active. The existing dep
- **Oracle-Gated Payment Channel Claims** (payments) Add optional sfOracleConditionID (Hash256, a PriceOracle keylet), sfConditionThreshold, and sfConditionOp (UInt8: GTE/LTE) to the PayChannel object, s

Appendix — commits

Time	Repo	Commit	Subject
01:44	xrpl-validator	cae6b85	fix(probe): a dropped fetch is not a divergence — withhold the verdict
09:46	xrpl-validator	44d7a1c	fix(ledger): a hydrated maker with no line for what it sells holds ZERO
14:47	xrpl-validator	f3f8b98	fix(xrpl-ledger): auth is not an interest in a trust line
17:48	xrpl-validator	dbb9110	feat(xrpl-ledger): DX_SEL — measure strand selection against rippled's model
18:41	xrpl-validator	005ad9a	fix(xrpl-ledger): the bridge strand's upper bound must not charge the transfer fee
19:52	xrpl-validator	03c2cb9	fix(xrpl-ledger): strand selection — upper-bound order with fall-through
21:40	xrpl-validator	a836f94	fix(xrpl-ledger): a rejected candidate's offer removals must survive its rollback